

MISSOURI



MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 3.2

Properties of Determinants

Row Operations and Determinants

Let A be a square matrix.

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$.

Example 1

Find the determinants.

$$\begin{vmatrix} -2 & -2 & 2 \\ -6 & -5 & 5 \\ -6 & 9 & 15 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 3 & -3 \\ 3 & 4 & -4 \\ 2 & -3 & -5 \end{vmatrix}$$

Example 2

Find the determinant.

$$\begin{vmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{vmatrix}$$

Row Operations and Determinants

Suppose a square matrix A has been row reduced to an echelon form U by replacements and interchanges only.

If there are r interchanges, then

$$\det A = (-1)^r \det U$$

Since U is in echelon form, it is upper triangular, and thus $\det U$ is the product of the diagonal entries.

If A is singular, at least one row of U will be all zeros, so there will be at least one zero on the diagonal and thus $\det U$ (and $\det A$) will be zero.

Numerical Note

Evaluation of an $n \times n$ determinant using row operations requires about $2n^3/3$ arithmetic operations.

A modern computer can calculate a 25×25 determinant in a fraction of a second, since only about 10,000 operations are required.

Theorems

Let A and B be $n \times n$ matrices. Then,

- A is invertible if and only if $\det A \neq 0$.
- $\det A^T = \det A$.
- $\det AB = (\det A)(\det B)$.

Example 3

Let A and B be square matrices. Show that even though AB and BA may not be equal, it is always true that $\det AB = \det BA$.

Example 4

True or False.

If the columns of a square matrix A are linearly dependent, then $\det A = 0$.
